Edwina Aylward

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Research Interests

Arithmetic geometry; elliptic and hyperelliptic curves; parity conjectures; local Galois representations.

Education

University College London 2023–2027 (expected)
PhD in Mathematics, London School of Geometry and Number Theory *London, UK*
Supervisor: Vladimir Dokchitser. EPSRC-funded doctoral student.

University of Cambridge 2022–2023
Master of Advanced Study in Mathematics (Part III), Trinity College *Cambridge, UK*
Distinction (88%); ranked 20th of 282 students.

Trinity College Dublin 2018–2022
B.A. (Mod) in Mathematics *Dublin, Ireland*
First-class honours (92%); first in class; Gold Medal for academic merit.

Research Papers

- **E. Aylward**, "On the integral 2-adic Tate module of elliptic curves", arXiv:2608.17725, 2026.
- **E. Aylward**, L. Cowland Kellock, V. Dokchitser and E. Lupoian, "Reduction Types of Genus 2 Curves", arXiv:2607.07558, 2026.
- **E. Aylward**, "Tamagawa numbers and positive rank of elliptic curves", *Mathematical Proceedings of the Cambridge Philosophical Society* (2026), doi:10.1017/S0305004126101947.
- **E. Aylward**, V. Dokchitser, H. Green and A. Morgan, "The parity conjecture for hyperelliptic curves", in preparation.

Invited Talks

- *Tamagawa numbers and positive rank of elliptic curves. "EnCounteRs with Don"* ECR event, Durham University (June 2026).
- *On the integral* 2*-adic Tate module of elliptic curves.* Tamagawa numbers of curves meeting, Malvern (April 2026).
- *ℓ-parity results over global function fields.* London Junior Number Theory Seminar, King's College London, and Linfoot Number Theory Seminar, University of Bristol (November–December 2025).
- *p-parity conjecture for twists of hyperelliptic curves by Artin representations.* Manchester Number Theory Seminar, University of Manchester (November 2025).
- *Rank predictions: parity vs. L-value approaches.* Junior Number Theory Seminar, University of Warwick (November 2024).

Contributed Talks and Posters

- *Reduction types of genus 2 curves (poster).* Regensburg GAP Days 1, University of Regensburg (July 2025).
- *Reduction types of genus 2 curves.* LMFDB, Computation, and Number Theory (LuCaNT) 2025, ICERM, Providence, Rhode Island (July 2025).
- *Reduction types of genus 2 curves.* Women in Number Theory and Geometry Spring Retreat (WINGs), Haydock (April 2025).
- *Elliptic curves with positive rank.* Elliptic Curves in the Cotswolds, Stonehouse (March 2025).
- *Artin formalism and BSD.* Young Researchers in Algebraic Number Theory VI, University of Oxford (August 2024).

Study Group Talks

- *Hilbert's Theorem 94 and the capitulation kernel.* Global Class Field Theory Study Group, UCL (January 2026).
- *Tate modules of hyperelliptic curves.* Genus 2 and Hyperelliptic Curves Study Group, UCL (November 2025).

- *Explicit regular models of hyperelliptic curves.* Models of Curves and Arithmetic Applications Study Group, UCL (March 2025).
- *Extended Weyl groups and affine Dynkin diagrams.* Bruhat–Tits Theory Study Group, King's College London (November 2024).
- *Artin's primitive root conjecture.* Open Problems in Number Theory Study Group, UCL (November 2024).
- *Introduction to étale cohomology.* Galois Cohomology Study Group, King's College London (May 2024).
- *Ray class fields.* Class Field Theory Study Group, UCL (April 2024).

Organisation

- Organiser, Surfaces Study Group, UCL, May–June 2026: eight-week study group on the arithmetic and geometry of algebraic surfaces.
- Co-organiser, Women in Number Theory and Geometry Spring Retreat (WINGs), 13–16 April 2026.
- Co-organiser, Elliptic Curves in the Cotswolds, 3–4 March 2025.
- Co-organiser, London Junior Number Theory Seminar, 2024–2025.

Teaching Experience

Tutorials and grading in Algebra II and Number Theory (UCL); tutorials and grading in Metric Spaces and Topology, Linear Algebra I, and Introduction to Abstract Algebra (King's College London); tutorials in Combinatorics (Trinity College Dublin).

Scholarships and Prizes

- **Trinity Studentship in Mathematics** (2022), Trinity College, Cambridge: studentship covering course fees and living expenses for the MASt.
- **George Moore Scholarship** (2022): scholarship support for postgraduate study; partial funding accepted alongside the Trinity Studentship.
- **Bishop Law Prize** (2022), Trinity College Dublin: awarded for graduating first in class.
- **Hamilton Prize** (2021), awarded by the Royal Irish Academy.

Conferences and Workshops Attended

- Young Researchers in Algebraic Number Theory, University of Bristol, September 2026.
- CAVARET 2: Curves, Abelian Varieties and Related Topics, University of Barcelona, July 2026.
- Seventeenth Algorithmic Number Theory Symposium (ANTS XVII), University of Groningen, July 2026.
- *"EnCounteRs with Don"* ECR event, Durham University, June 2026.
- Women in Number Theory and Geometry (WINGs), Cheltenham, UK, April 2026.
- Tamagawa numbers of curves meeting, Malvern, UK, April 2026.
- Algebraic and geometric methods for Diophantine problems, Pisa, September 2025.
- Regensburg GAP Days 1, University of Regensburg, July 2025.
- LMFDB Workshop, Massachusetts Institute of Technology, July 2025.
- LMFDB, Computation, and Number Theory (LuCaNT), ICERM, Providence, Rhode Island, July 2025.
- Women in Number Theory and Geometry (WINGs), Haydock, UK, April 2025.
- Elliptic Curves in the Cotswolds, Stonehouse, UK, March 2025.
- Young Researchers in Algebraic Number Theory, University of Oxford, August 2024.

Computational Projects

- Contributed local root number data for genus 2 curves to the L-functions and Modular Forms Database (LMFDB).
- Developed the companion website for *Reduction Types of Genus 2 Curves*.